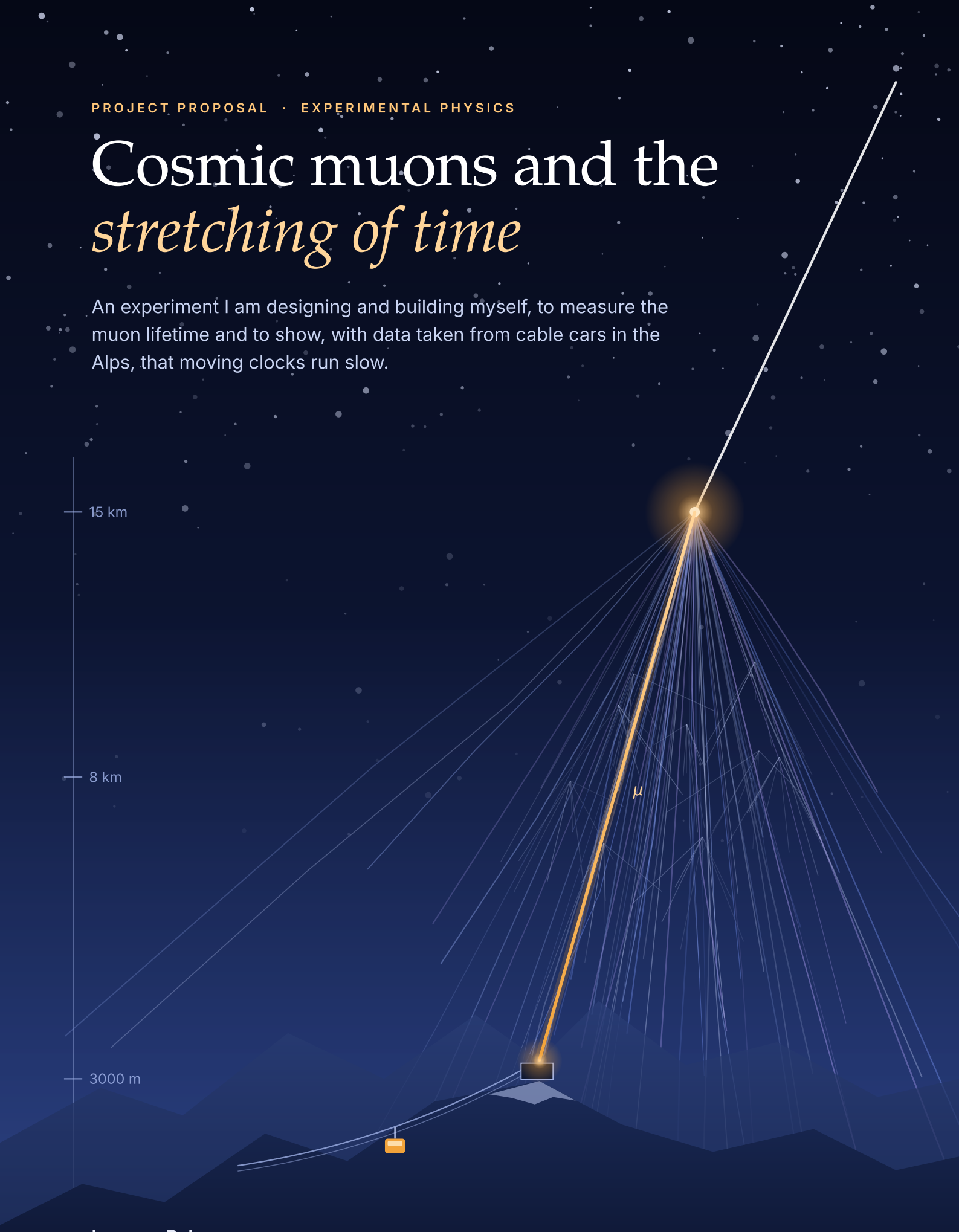
PROJECT PROPOSAL · EXPERIMENTAL PHYSICS

Cosmic muons and the *stretching of time*

An experiment I am designing and building myself, to measure the muon lifetime and to show, with data taken from cable cars in the Alps, that moving clocks run slow.



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PROJECT OVERVIEW
Version 1.0 · November 2026

ABSTRACT

Muons are created about 15 km above our heads and live, on average, only 2.2 millionths of a second. Without special relativity almost none of them would reach the ground; instead about one per square centimetre per minute does. In this project I build, from scratch and on a student budget, a detector that measures the two halves of that story: the muon lifetime at rest, in a plastic scintillator at home, and the change in muon flux between Milan and 3000 m, carried up the mountain by cable car. Comparing the two with a simulation tests time dilation directly.

WHAT

A two-part measurement of relativistic time dilation with cosmic-ray muons, with an apparatus I design, build and analyse myself.

WHERE IT STANDS

Physics, electronics, firmware and mechanics are designed and simulated. The custom timer already works on real hardware.

ESTIMATED COST

About **€575–860** for the complete apparatus, 40–50% of it for the plastic scintillators. Tools are borrowed and all software is free.

2.2 µs
muon lifetime at rest, target precision 2–3%

×95 vs <×2
flux increase at 3000 m without and with relativity

~745
usable muon decays per day in the 1-litre detector (simulated)

13.3 ns
time resolution of the custom timer, verified on hardware

Why this project

Time dilation is usually met as a formula on a blackboard. Here it becomes a number read off a histogram, from an instrument whose every part — scintillator, light sensor, front-end amplifier, timing firmware and analysis — I have designed myself and can explain. The project is deliberately complete: it covers detector physics, analogue and digital electronics, mechanical design, Monte Carlo simulation and statistical analysis, the core skills of experimental particle physics.

It is also deliberately cheap and open. All schematics, circuit boards, 3D models, firmware, simulations, raw data and analysis code will be published in a public repository, so that other students can rebuild the setup or use it in teaching labs and outreach.

How this document is organised

Section 1 explains the physics and Section 2 the two measurements and how I will judge them. Section 3 describes the apparatus, Section 4 what I have built and tested so far, and Section 5 what the simulations predict. The last sections cover the work plan and risks, the estimated cost, and the references.

SECTION 1

1 The muon paradox

Particles that should not reach the ground, and yet do.

Cosmic rays, mostly protons from outside the solar system, hit the upper atmosphere and produce showers of secondary particles. Among them are **muons**, heavier cousins of the electron. A muon at rest decays into an electron and two neutrinos with a mean lifetime of $\tau = 2.197\ \mu s$. Even travelling at almost the speed of light, in that time it would cover on average only $c\tau \approx 659$ m.

Muons are born around 15 km up. If their clocks ran like ours, their number would drop by a factor e for every 659 m of descent, and the ratio between the flux at 3000 m and on the plain would be about **95**. Measurements show a ratio of **less than 2**. Special relativity explains the difference: for a muon of a few GeV the Lorentz factor $\gamma = E/mc^2$ is about 40, so in our frame its lifetime is stretched forty times and its decay length grows to tens of kilometres.

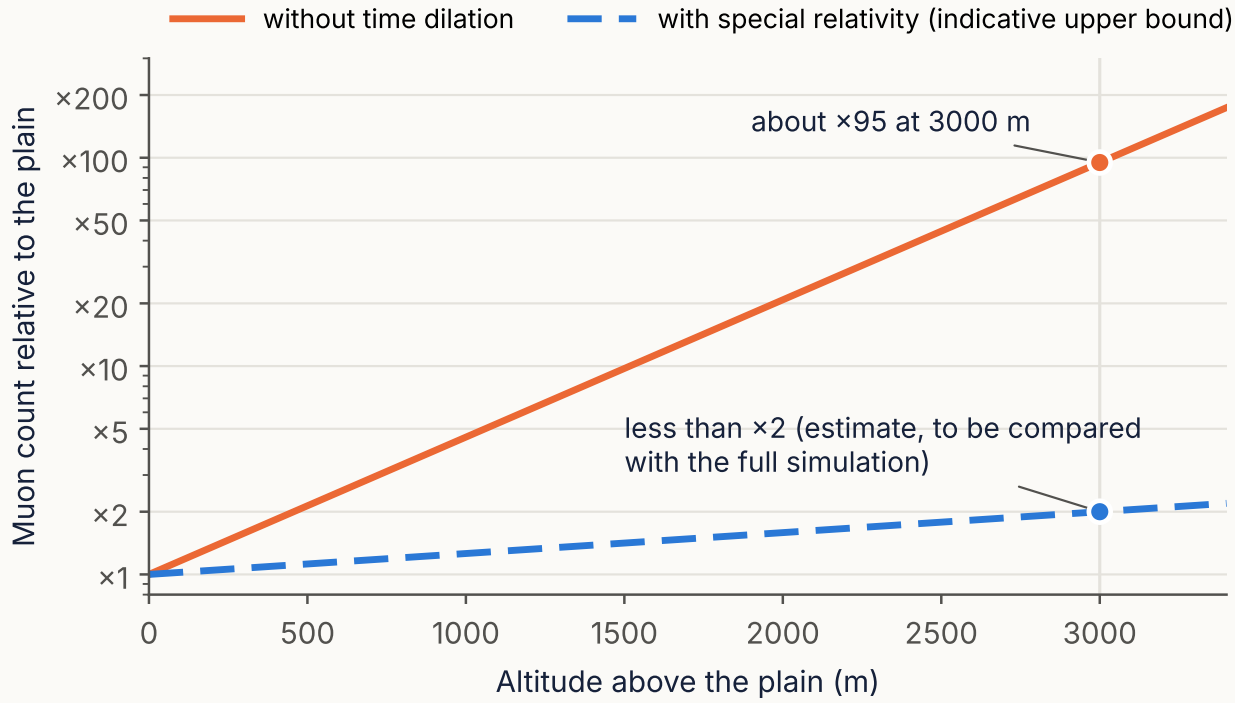


Figure 1. Muon count at altitude relative to the plain. Orange: what non-relativistic decay would predict, exp(h / cτ). Blue: the order of magnitude expected with relativity (less than ×2 at 3000 m), drawn as an indicative upper bound; the real comparison will use a simulation that includes the energy spectrum and energy loss in air.

> *The experiment measures both numbers with the same hands: the clock of a muon at rest, and how many muons survive the trip.*

The two measurements are complementary. The lifetime τ is measured *at rest*, with muons that stop inside the detector, where no relativity is involved. The flux at different altitudes is then predicted twice from that same τ — once with γ and once with $\gamma = 1$ — and compared with what the detector counts on the mountain. The data should fall on the relativistic curve and far below the classical one.

SECTION 2

2 Goals and success criteria

One result that stands on its own, and one that completes the story.

MEASUREMENT 1 · AT HOME

Muon lifetime at rest

A one-litre block of plastic scintillator stops about a thousand muons a day. Each stop gives a flash of light, and the decay electron a second one microseconds later. The distribution of those delays is an exponential whose slope is the lifetime.

Target: $\tau \approx 2.2$ μs with 2–3% total error, compared with the Particle Data Group value.

MEASUREMENT 2 · IN THE ALPS

Muon flux versus altitude

Two small scintillator tiles in coincidence, with a lead absorber between them, form a "telescope" that counts muons. I carry it by cable car and chairlift to 5–6 stops, from Milan up to 2500–3400 m, in winter 2027.

Target: 5–8% error per point; agreement with the relativistic model within 10–20%.

Final comparison

I run a Monte Carlo simulation of the muon flux through the atmosphere twice, using the lifetime from measurement 1: once with time dilation and once without. I then compare the measured ratios to the plain with both curves, and with published references: the EXPACS/PARMA atmospheric model, the BESS spectrometer data at 30 m and 2770 m, and the cosmic-ray review of the Particle Data Group. I use ratios instead of absolute rates, so that the detector acceptance cancels.

How I will judge success

CRITERION	TARGET	STATUS
Bias of the lifetime analysis	below 0.5%	PASSED IN SIMULATION
Correct statistical errors	pull width between 0.9 and 1.1	PASSED IN SIMULATION
Statistical error on τ	2% within 5 days, 1% within 3 weeks (apparent lifetime)	PASSED IN SIMULATION
Background from random pairs	below 5% of the signal in the first 5 μs	PASSED IN SIMULATION 0.65%
Timer: no lost events, correct delays	every delay within one clock step	PASSED ON HARDWARE
Lifetime from real muons	2–3% total error, consistent with PDG	MEASUREMENT 1
Flux ratios at altitude	on the relativistic curve within 10–20%	MEASUREMENT 2

Deliverables

- A short paper on the apparatus, the data, the systematic uncertainties and the result.
- A public repository with every design file, the firmware, the simulations, the raw data and the analysis, so that anyone can reproduce the result or rebuild the detector.
- A working, documented detector that can be reused for teaching or outreach.

SECTION 3

3 How the experiment works

From a flash of light to a timestamp with 13-nanosecond steps.

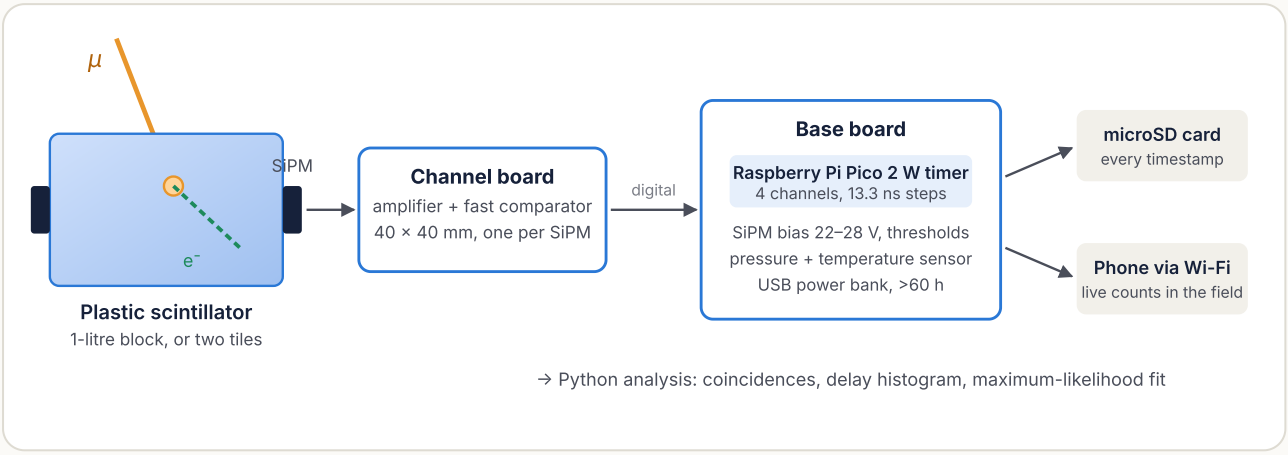


Figure 2. Signal chain. Light from the scintillator is read by silicon photomultipliers (SiPMs), which need no high-voltage supply. Each SiPM has its own small board that turns the pulse into a clean digital edge; the base board timestamps every edge and stores it.

Measurement 1: two flashes

A slow muon that stops in the block gives a bright flash. When it decays, the electron gives a second, smaller one. The timer records both; my analysis keeps pairs seen by *both* SiPMs within 55 ns, which removes the sensors' own noise, and fits the delays between 0.5 and 20 µs.

About four in ten stopped muons are negative, and inside the plastic they live slightly less, because some are captured by carbon nuclei. The fit therefore uses two exponentials, so that the free-muon lifetime can be compared with the reference value.

Measurement 2: a muon telescope

The same two channel boards move to two 5 × 5 × 1 cm tiles, one above the other with lead in between. Only particles that cross both tiles, and the lead, are counted; electrons and photons from the shower are stopped. A few hundred coincidences in half an hour give each altitude point.

I log the air pressure at every stop, so that altitude is expressed as the amount of air above the detector, the quantity that actually controls muon survival.

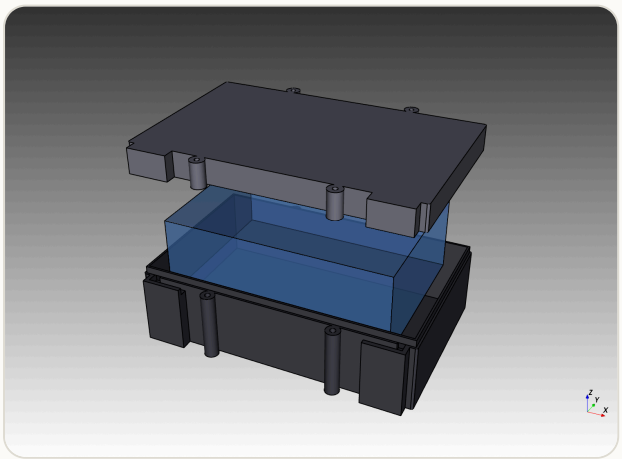


Figure 3. Light-tight enclosure of the 1-litre scintillator block (blue), which I designed in Onshape, with labyrinth seal and SiPM supports. To be 3D-printed in resin.

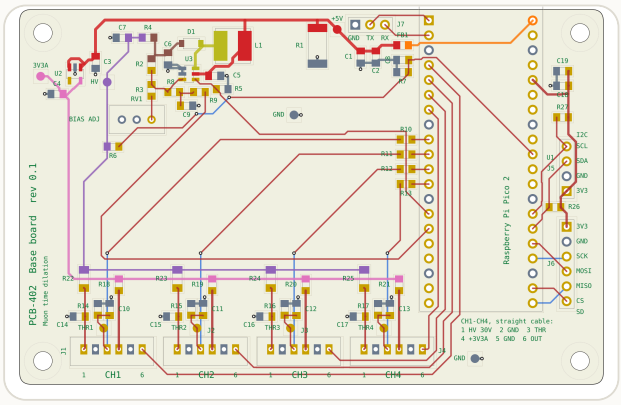


Figure 4. Layout of the base board (100 × 64 mm), which I designed in KiCad: SiPM bias supply, threshold settings, four channel connectors and the Pico timer.

SECTION 4

4 What I have done so far

Status in November 2026: from an idea to a design simulated end to end.

- ✓ **Physics and architecture**
  Measurement plan, detector choice, suppliers and costs; written introduction to the physics.
- ✓ **Simulation of the full apparatus**
  Muons in the block, light collection, analogue electronics (SPICE), timer and a complete pseudo-experiment, before buying anything.
- ✓ **Electronics designed**
  Channel board and base board in KiCad, design-rule check with zero errors, values tuned from simulation.
- ✓ **Boards ready for production**
  Manufacturing files and bill of materials for both boards, with parts chosen from distributor stock.
- ✓ **Timer firmware working**
  Written in C on the Pico's programmable I/O and tested on real boards: no lost events, correct delays.
- ✓ **Mechanics designed**
  Enclosures for the block and the tiles in CAD, checked for fit and clearances; print files exported.
- … **External design review**
  I have shared the schematics and board layouts with a professor for review.
- ○ **Detector materials**
  Plastic scintillators and SiPM sensors, to be purchased.
- ○ **Assembly and testing**
  Soldering, bring-up with the oscilloscope, light-tightness tests.
- ○ **Measurements and paper**
  Lifetime, altitude campaign, analysis, publication.

The timer, tested on real hardware

The most delicate digital part is the timer, which must timestamp pulses a few microseconds apart without missing any. I tested it on 13 October 2026 with a second Pico acting as a pulse generator that fires pairs of pulses with random, exponentially distributed delays of known mean 2.197 µs, a dress rehearsal of measurement 1.

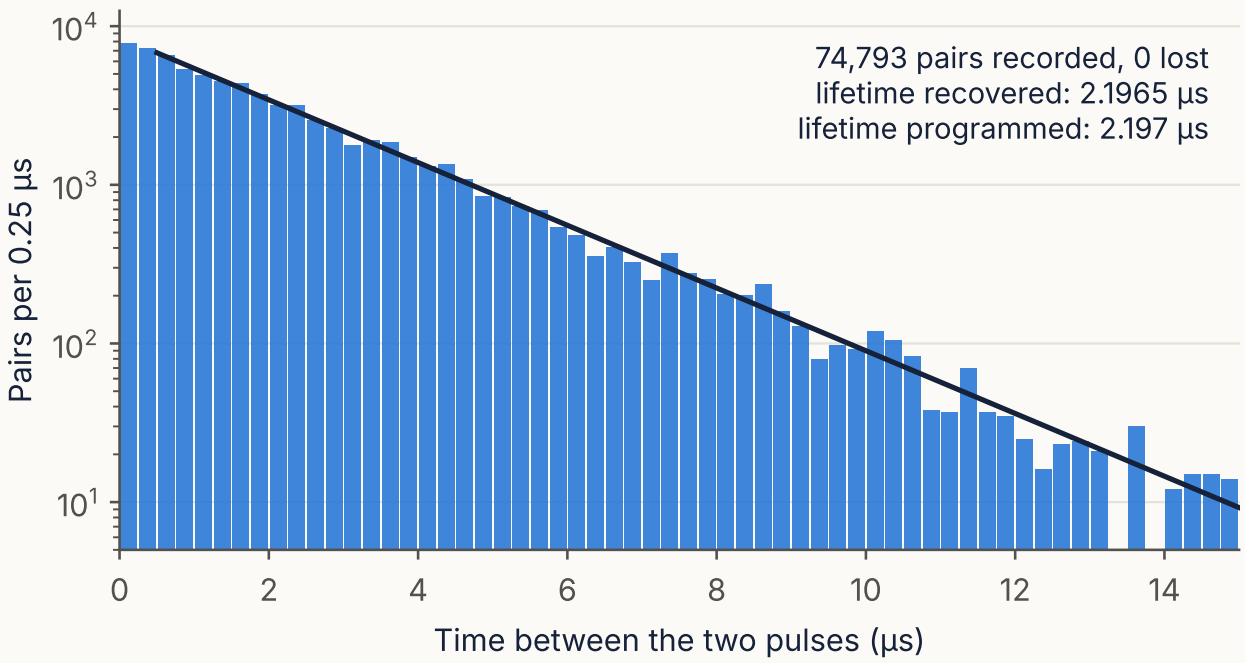


Figure 5. Delays measured by the timer in a 60-second test with the pulse generator (raw data, not muons). The fitted lifetime is 2.1965 µs against 2.197 µs programmed; the generator repeats a sequence of 10,000 delays, so the realistic uncertainty is about ±0.025 µs. Every one of the 74,793 measured delays matches the delay actually sent within one clock step.

SECTION 5

5 Expected results, from simulation

I ran the measurement hundreds of times on the computer before buying a single sensor.

Before ordering parts, I simulated the whole apparatus in five linked steps: muons crossing and stopping in the block (S1), light reaching the two SiPMs (S2), the analogue electronics in SPICE (S3), the timer (S4), and finally hundreds of complete pseudo-experiments analysed with the same script that will read the real data (S5). The simulation changed the design. It showed that the circuit of the reference open-source detector, used as it is, would hide about 40% of the decays behind a long pulse; I then tuned the new front-end in SPICE, changing three resistor values, until the dead time after a muon came down to 250 ns.

~1000
muons stopping in the block per day; ~980 decay with a visible electron (S1)

96.5%
detection efficiency for decays in the fit window, including dead time (S5)

0.65%
random background under the signal in the first 5 µs (S5)

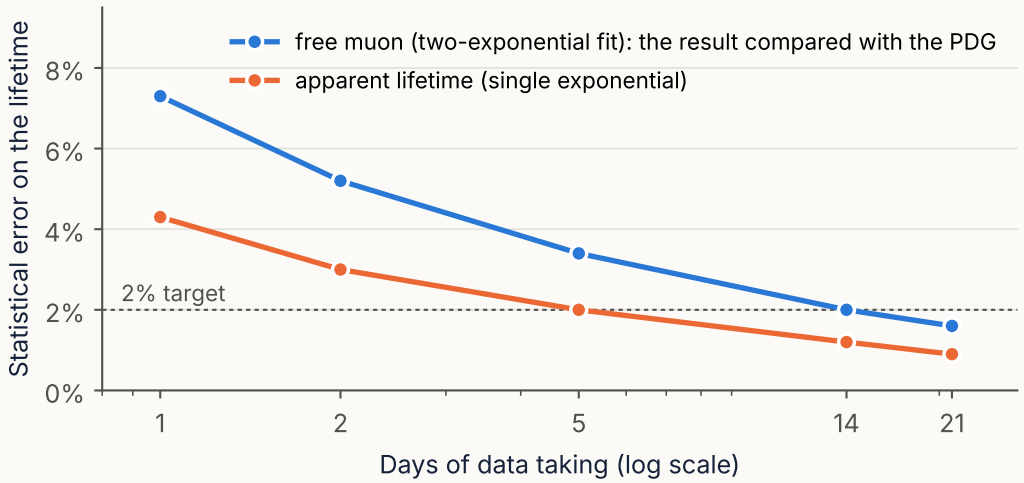


Figure 6. Statistical error on the lifetime versus days of data taking, from 60–500 simulated experiments per point. The free-muon lifetime, the number compared with the reference value, reaches 2% in about two weeks of continuous running; the detector runs unattended, so this costs time, not effort.

What the simulation says about the result

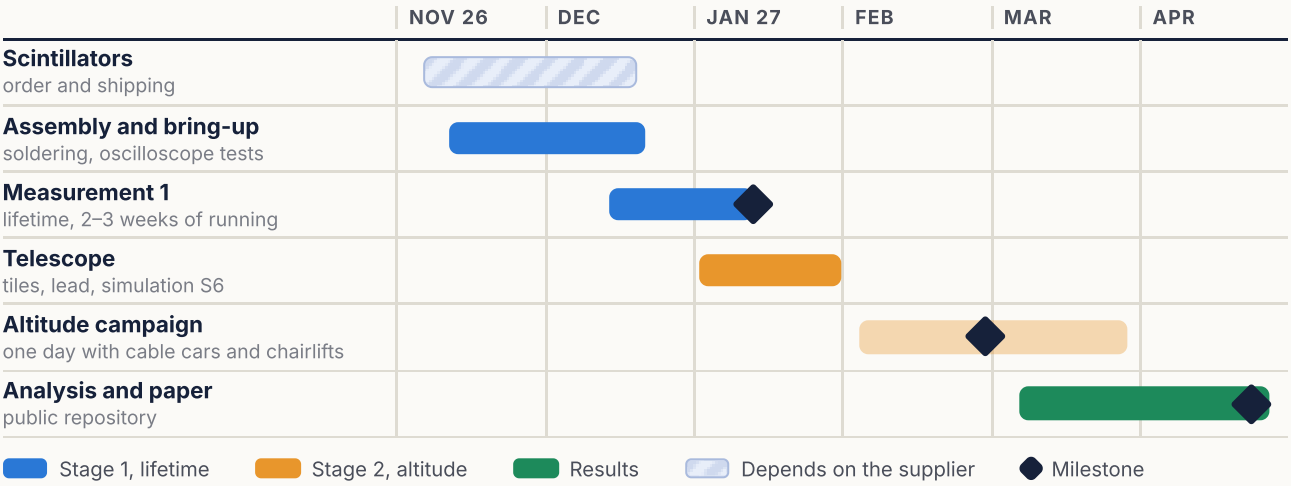
QUANTITY (5 DAYS, 500 PSEUDO-EXPERIMENTS)	VALUE	MEANING
Usable decay pairs per day	745	out of 772 an ideal detector would see
Bias of the two-exponential fit	+0.06%	the chain does not distort τ
Pull width	1.02	quoted errors are the true errors
Main systematic: μ^+/μ^- charge ratio	−0.4% to +0.1%	for a ratio between 1.10 and 1.30; small against statistics
Error after 14 days, free muon	2.0%	meets the 2–3% target

The simulation let me find and fix two analysis mistakes before any real data: a fit window edge falling exactly on a multiple of the timer step (−0.5% bias) and a normalisation error in the two-exponential fit. Not yet simulated: the telescope and the altitude campaign (step S6), which I will do before winter.

SECTION 6

6 Work plan and risks

Two stages, part-time alongside my university courses.



Indicative dates. Measurement 1 starts as soon as the scintillators arrive; the altitude day has a wide February–March window chosen around the exam session and the weather.

Main risks and how they are handled

RISK	WHAT WOULD HAPPEN	MITIGATION
Scintillator supply	Measurement 1 starts later	Several suppliers in Europe and Asia identified; a bar-shaped block of similar volume was simulated and collects light just as well, so the shape can adapt to what is available.
SiPM soldering fails	A channel does not work	Five copies of each board; a practice board first; a solder-paste stencil for the base board.
Light leaks into the enclosure	Rates jump with room light	Labyrinth seal, black tape, bias only with the box closed; a torch test before every run.
After-pulses distort the exponential	Excess at short delays	Two-SiPM coincidence, lower time cut, check that τ is stable when the cut moves.
Cold at altitude	SiPM gain drifts with temperature	Temperature logged at every stop; bias compensation prepared on the board; power bank tested in the cold.
Telescope too heavy for a cable car	Impractical to carry	A stack of 1-mm lead sheets instead of a lab brick.

SECTION 7

7 Estimated cost

A complete particle detector for the price of a mid-range phone.

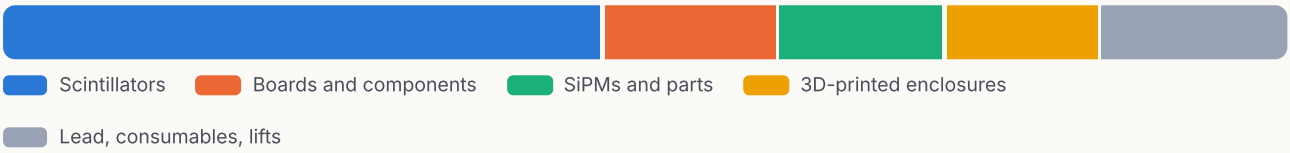
COMPLETE APPARATUS

€575–860

estimated total for both measurements, VAT and shipping included. The range comes almost entirely from the scintillators, whose price depends on the supplier.

NOT IN THE TOTAL

Oscilloscope, soldering station and hand tools, which I borrow from a school laboratory, and software: KiCad, Onshape, ngspice and Python are all free.



#	ITEM	LOW	HIGH
1	**Plastic scintillators**: one ~1-litre block for measurement 1, two 5 × 5 × 1 cm tiles for the telescope Range from prices listed or quoted by several suppliers, shipping and VAT included.	€225	€450
2	**Printed circuit boards**: two designs, five copies each, solder-paste stencil (JLCPCB)	€14	€14
3	**Electronic components** for both boards (LCSC)	€69	€69
4	**Two Raspberry Pi Pico 2**, for timer tests and as a pulse generator	€13	€13
5	**Sensors and parts** (Mouser): two onsemi MicroFC-60035 SiPMs, Raspberry Pi Pico 2 W, pressure and temperature sensor, microSD module, optical grease, screws, cables, flux	€92	€92
6	**3D-printed enclosures** in resin (JLC3DP), seven parts with threaded inserts Range includes possible import charges.	€75	€95
7	**Consumables**: light-tight tape, gloves, gaskets, springs, matt black paint	€18	€18
8	**Lead absorber** for the telescope, 1-mm sheets	€27	€27
9	**Altitude campaign**: lift tickets for one day Depends on the route.	€40	€80
	Estimated total	**€573**	**€858**

All figures are estimates from prices read in October 2026, VAT included; US-dollar amounts converted at about €0.89. The colour bar uses the middle of each range.

SECTION 8

8 Outputs, about me and references

What remains once the measurement is done.

WHAT WILL BE PUBLISHED

- A short paper with the method, the data, the systematic uncertainties and the result.
- Every design file: schematics, circuit boards, 3D models and the bill of materials.
- The firmware, the simulations, the raw data and the analysis code.
- Notes from each milestone: first muon pulses, first lifetime fit, altitude day.

WHAT I WANT TO LEARN

- How a complete experiment works, from the first design choice to the systematic errors.
- Detector physics with scintillators and SiPMs, the same technology used in many particle-physics experiments.
- A solid starting point for a future in experimental particle physics.

About me

I am **Lorenzo Bologna**, a second-year student of Physics Engineering (Ingegneria Fisica) at Politecnico di Milano, with experience in mechanical CAD and in computing. This is a personal project that I run on my own: the design, the construction, the measurements and the analysis are mine. I want to become a research physicist in experimental particle physics.

Related work

My detector follows the approach of **CosmicWatch**, an open-source desktop muon detector based on scintillators and SiPMs, but I redesigned the front-end for decay timing, replaced its single-channel readout with a four-channel timer, and added the altitude measurement. Classic precedents are the Rossi–Hall measurement of 1941 and the Frisch–Smith film *Time Dilation: An Experiment with Mu-Mesons* (1963), which compared muon rates on Mount Washington and at sea level.

References

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Thank you for reading.
Questions, comments and suggestions are very welcome.

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